

Degradation phenomena and degradation mechanisms for highly conductive PEDOT:PSS films

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ABSTRACT

Poly (3, 4 – ethylenedioxythiophene) – poly (styrenesulfonate) (PEDOT: PSS) thin film has caught much attention because of its high electrical conductivity, optical transmittance and mechanical flexibility. Reducing the PSS content can improve the conductivity of PEDOT: PSS film effectively. However, few studies have been reported on the corresponding degradation behaviors and mechanisms for the PEDOT: PSS films with different PSS contents. Here, we studied the degradation phenomena under various aging conditions. Our results indicated that the degradation behaviors are determined by the content of PSS, and the degradation mechanism is related to the change of molecular configuration of PEDOT chains under various aging conditions.

1. Introduction

Transparent conductive materials have been widely used for a long time [1]. ITO (Indium tin oxide) is a common transparent conductive material. However, ITO is expensive and lacks of mechanical flexibility. Therefore, low-cost and flexible transparent conductive materials are needed. PEDOT can be a potential transparent conductive material because of its high electrical conductivity, excellent transmittance, high mechanical flexibility and simple preparation process [2–4]. However, for the pristine PEDOT: PSS film, the electrical conductivity is about 1 S/cm [2]. To improve the conductivity, much work has been reported, for example, the conductivity values over 1000 S cm⁻¹ were obtained by the organic solvents and more than 2000 S cm⁻¹ by adding ionic liquids [3]. It is worth noting that, the H₂SO₄ post-treatment method found by Kim *et al.* in 2014 made the conductivity of PEDOT increase to 4380 S cm⁻¹ [2].

Because PEDOT is hydrophobic, the hydrophilic PSS acts as a soluble template for solution-prepared PEDOT: PSS films. A certain content of PSS in solution is necessary to deposit PEDOT: PSS film by spin-coating. For the commercial PEDOT: PSS solutions, the mass ratio of PEDOT: PSS for Clevios P VP CH 8000, Clevios P VP AI 4083, and Clevios pH 1000 is 1/20 [5], 1/6 [6] and 1/2.5 [7], respectively. After PEDOT: PSS film was formed, it always prefers a phase segregated structure consisting of PEDOT grains surrounded by a shell formed by PSS [8]. Because PEDOT is conductive while PSS is non-conductive, removing excessive PSS

would be helpful to increase the conductivity.

Recently, we developed a two-step method for H₂SO₄ post-treatment and the conductivity of 6323.9 ± 364.5 S/cm was obtained for PEDOT: PSS film [9]. As we know, it is the highest conductivity of PEDOT prepared by solution method. Our method for achieving high conductivity is simple and easy to realize in industry. It only includes low concentration and high concentration H₂SO₄ treatments. The removal of PSS and redox reaction are the keys to enhance the conductivity [9]. However, in the case for lacking of PSS, the electrical stability of PEDOT: PSS film is worthy to be studied further. It is particularly important for device application, while there are few reports on it [10–11]. In this paper, we investigated the aging phenomena and the aging mechanisms for PEDOT: PSS films with different PSS contents.

2. Experimental

PEDOT: PSS (Clevios pH 1000, mass ratio 1/2.5, solid content 1.0–1.3%) and H₂SO₄ (Analytical purity, 95–98%) were supplied by Xi'an polymer Light Technology Cory and BeiJing JOHN LUNDA technology development CO., respectively. PEDOT: PSS films were obtained by spin-coating on glass substrates. The speed and the time of spin-coating are 2000 rpm and 60 s. To remove the excessive PSS, the samples were immersed into H₂SO₄ with various concentrations for 10 min. Two series of films were prepared: (1) the films treated by H₂SO₄ once. The H₂SO₄ concentrations were 9.2 and 18.4 mol/L, so we named the

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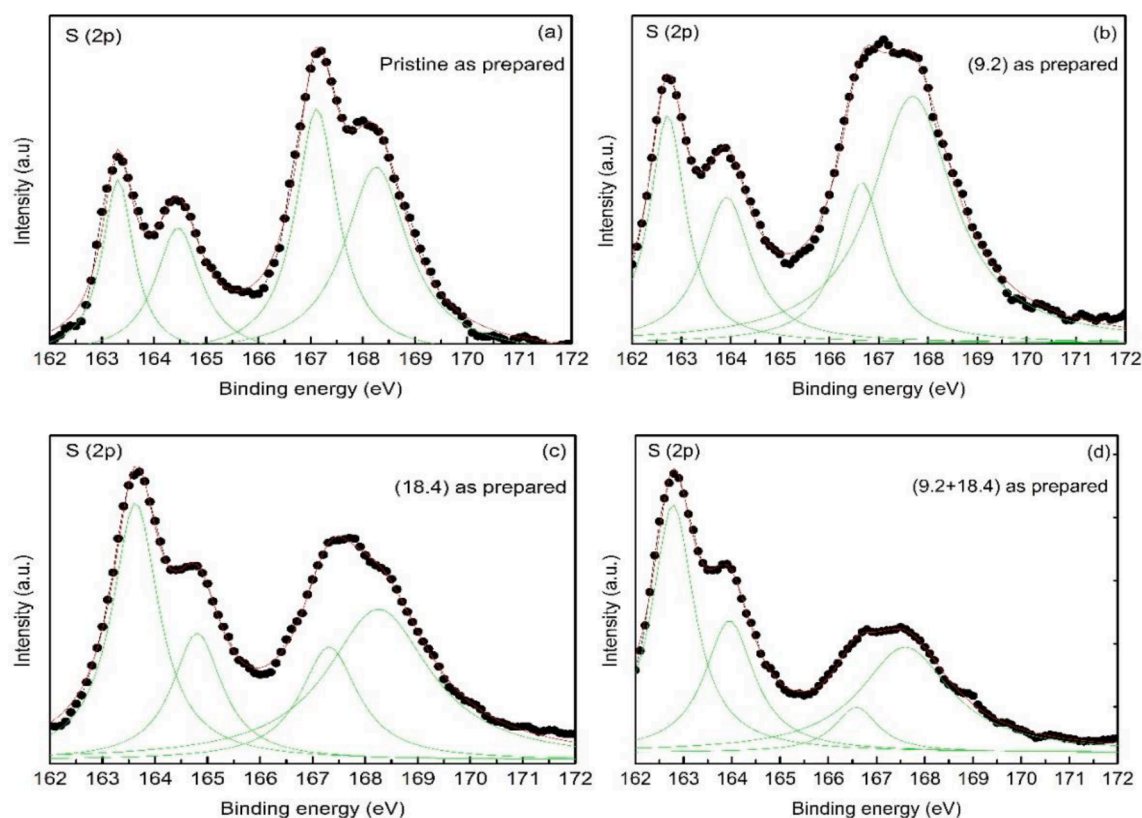


Fig. 1. S (2p) signals for the PEDOT: PSS films as prepared.

Table 1

$R_{PE/PS}$, sheet resistance, thickness and conductivity for the PEDOT: PSS films as prepared. The conductivity σ was obtained by $\sigma = 1/RT$ (where R is sheet resistance and T is film thickness).

Samples	Pristine	(9.2)	(18.4)	(9.2+18.4)
$R_{PE/PS}$	0.51	0.56	0.83	1.47
Sheet resistance (Ω/sq)	20047.8	156.3	91.0	71.0
Thickness (nm)	120.1	73.4	46.1	26.3
Conductivity (S/cm)	4.2	871.7	2383.7	5355.3

films as the (9.2) and (18.4) film, respectively. (2) The films treated by H_2SO_4 twice. The H_2SO_4 concentrations were 9.2 and 18.4 mol/L, so the film was named as the (9.2+18.4) film. Sheet resistances were measured by a four-point probe system (RTS-8, Guangzhou four probe technology Co., Ltd). For each film, in the area of 2 cm $\times$ 2 cm, there are equal spacing 9 points in 3 rows and 3 columns, which are set as the measured points for sheet resistance measurements. The standard deviation is about 4%. Film thicknesses were obtained by an alpha step surface profiler (Veeco Dektak 150). X-ray photoelectron spectroscopy (XPS) were measured by a Kratos-Axis Supra spectrometer (Al-K, 1486 eV). Raman spectra were obtained by a spectrometer (Renishaw In Via, 532 nm).

3. Results and discussion

Quantitatively, XPS was used to estimate PSS extraction by acid treatment. Because the S atom of PSS is inside the sulfonate moiety and

that of PEDOT is located at the thiophene ring, the binding energies of S (2p) electrons are different, such as: S $2p_{1/2}$ (168–169 eV) and S $2p_{3/2}$ (167–168 eV) for PSS, while S $2p_{1/2}$ (164–165 eV) and S $2p_{3/2}$ (163–164 eV) for PEDOT, reported by Bae [12] and Ling [13]. The S (2p) lines were curve-fitted by Origin program as presented in Fig. 1. The molar ratio of PEDOT/PSS ($R_{PE/PS}$) was obtained by the area ratio between the S (2p) signals from PEDOT and PSS, as listed in Table 1. For the pristine PEDOT: PSS (Clevios pH 1000), the mass ratio of PEDOT: PSS is 1/2.5, so $R_{PE/PS}$ is about 1/1.9, that is 0.53. The value of $R_{PE/PS}$ for our pristine film is 0.51, and increases as treated by H_2SO_4 . For the (9.2+18.4) film, $R_{PE/PS}$ reaches 1.47, indicating that it contains more PEDOT than PSS. Because PEDOT is conductive while PSS is insulating, and PEDOT grains are surrounded by PSS shells, the carriers will hop between PEDOT grains to form a conductive path [14]. Therefore, the (9.2+18.4) film has the highest conductivity among all the films as the hopping of charges between PEDOT grains is easiest. As listed in Table 1, the thickness of the pristine film is 120.1 nm and reduces gradually with

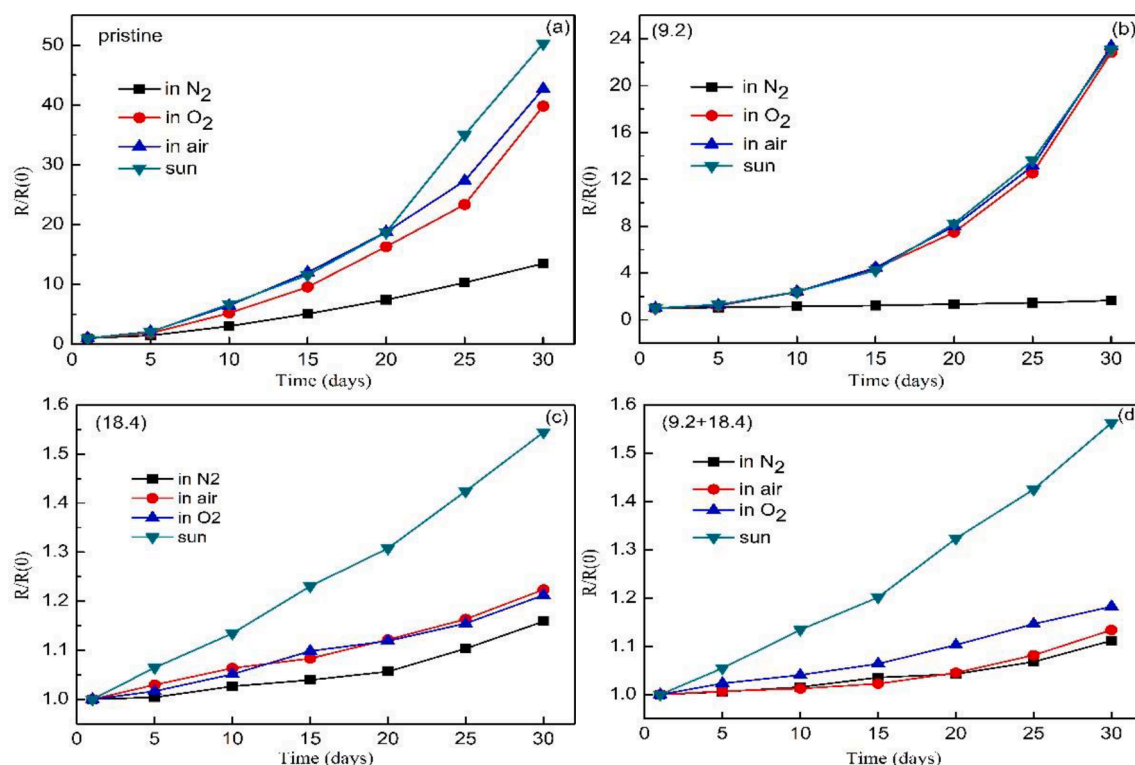


Fig. 2. Dependence of $R/R(0)$ on the aging time for PEDOT: PSS films under different aging conditions.

increasing H_2SO_4 concentration. For the (9.2+18.4) film, the thickness becomes 26.3 nm, which is about 1/4.5 of the original thickness. Table 1 also shows an obvious decrease of sheet resistance and a rapid increase in conductivity with increasing H_2SO_4 concentration. The conductivity is 5355.3 S/cm for the (9.2+18.4) film, which is higher than 4380 S/cm obtained by Kim who first found the H_2SO_4 treatment [2].

Here we focused on the degradation phenomena of PEDOT: PSS films. For aging, the films were placed under four different conditions at room temperature for 30 days: (1) dark, in N_2 (intrinsic aging); (2) dark, in O_2 (effect of O_2); (3) dark, in air (effect of $O_2 + H_2O$); and (4) outdoors in Beijing (effect of light + $O_2 + H_2O$). In the aging days (from May 9 to June 7, 2021), there were 4 sunny days, 21 cloudy days and 5 rainy days. The average temperature was about 22 °C and the average relative humidity was about 23%. To investigate the degradation phenomena, we calculated the $R/R(0)$ values, where $R(0)$ is the initial sheet resistance and R is the sheet resistance. As shown in Fig. 2, the pristine film has a rapid increase of square resistance with the aging time. While for the treated films, $R/R(0)$ becomes stable with H_2SO_4 concentration increasing, and is almost unchanged as the H_2SO_4 concentration is 18.4 mol/L. Either in O_2 or air, even under illumination, $R/R(0) < 1.6$ for the (18.4) and (9.2+18.4) films. It means that PSS extraction by H_2SO_4 treatment is helpful for electrical stability of PEDOT: PSS films.

It is easy to find that illumination is the main aging factor because $R/R(0)$ for all the films is sensitive to light. This result is consistent with the one Schultheiss reported [11]. It was reported that water from humidity significantly affects the conductivity of PEDOT [15]. For the pristine film and the (9.2) films, the effect of water on $R/R(0)$ is obvious. While for the (18.4) and (9.2+18.4) films, the effect of water on $R/R(0)$ is much less than that of light. The reason may be that, for PSS is

hydrophilic, the films with less PSS are thus less affected by water during aging.

Raman spectroscopy is powerful to study molecular structure of conducting polymers. Raman spectra for the as prepared films are shown in Fig. 3(a). The main peak of the pristine film appears at about 1440 cm^{-1} . This peak red-shifts regularly with H_2SO_4 concentrations increasing. This shift is related to the transform from benzoid structure to quinoid structure in PEDOT chains supposed by Ouyang *et al.* [16]. These two structures have different conjugated π -electrons. The benzoid prefers coil conformation structure which results in a low conductivity, while the quinoid is linear or expanded-coil structure and thus has a high conductivity [17]. Our results indicated that, both benzoid and quinoid structure may co-exist in the pristine film. After H_2SO_4 treatment, a part of benzoid structure was transformed to the quinoid structure. As treated by concentrated H_2SO_4 , the quinoid structure is the main structure for PEDOT chains, resulting in an enhancement of conductivity.

Under different aging conditions for 30 days, Raman spectra were measured again. As shown in Fig. 3(b and c), for the pristine film and the (9.2) film, the peak located at 1440 cm^{-1} has an obvious blue-shift after aged. It indicates that a transformation from the quinoid structure to the benzoid structure occurs during aging. Especially by the effect of light, this change is particularly obvious. However, for (18.4) and (9.2+18.4) films, the peak at 1440 cm^{-1} has a slight blue-shift even under sun illumination. It means that, the decrease of PSS makes the molecular structure dominated by quinoid structure, and maintains quinoid structure to a great extent under various aging conditions. This is a possible reason why the (18.4) and (9.2+18.4) films have electrical stability.

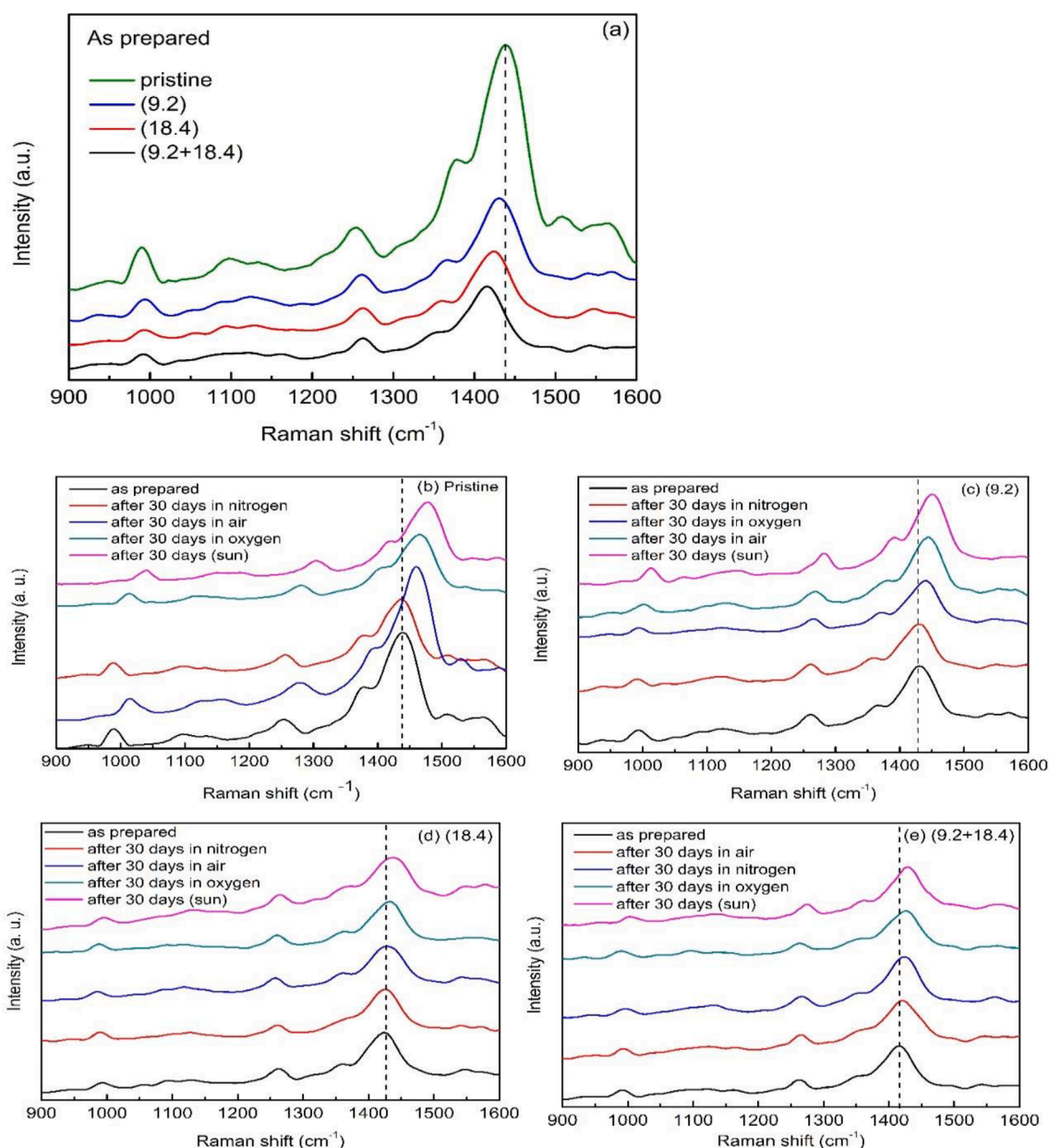


Fig. 3. Raman spectra for the films before and after aging.

4. Conclusions

PEDOT:PSS films were spin-coated and post-treated with H_2SO_4 of different concentrations. Degradation phenomena and mechanisms for the films with different PSS contents were studied under different aging conditions. Our results indicated that, degradation phenomena are related to the PSS content, and transformation of molecular configuration is the main aging reason. The (9.2+18.4) film has a least content of PSS and a high conductivity of 5355 S/cm, while the remaining PSS can keep the quinoid structure in PEDOT chains and shows a stable electrical performance under various aging conditions for 30 days. Our results are important for the practical application of PEDOT.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] D. Ginley, C. Bright, Transparent conducting oxides, *MRS Bull.* 25 (8) (2000) 15–18, <https://doi.org/10.1557/mrs2000.256>.
- [2] N. Kim, S. Kee, S. Lee, B. Lee, Y. Kahng, Y. Jo, B. Kim, K. Lee, Highly conductive PEDOT: PSS nanofibrils induced by solution-processed crystallization, *Adv. Mater.* 26 (2014) 2268–2272, <https://doi.org/10.1002/adma.201304611>.
- [3] H. Shi, C. Liu, Q. Jiang, J. Xu, Effective Approaches to Improve the Electrical Conductivity of PEDOT:PSS: A Review, *Adv. Electron. Mater.* 1 (2015) 1–16, <https://doi.org/10.1002/aelm.201500017>.
- [4] J. Thomas, L. Zhao, D. McGillivray, K. Leung, High-efficiency hybrid solar cells by nanostructural modification in PEDOT:PSS with co-solvent addition, *J. Mater. Chem. A* 2 (2014) 2383–2389, <https://doi.org/10.1039/C3TA14590E>.
- [5] M. Döbelin, R. Marcilla, C. Tollan, J.A. Pomposo, J.-R. Sarasua, D. Mecerreyes, A new approach to hydrophobic and water-resistant poly (3, 4-

- ethylenedioxythiophene): poly (styrenesulfonate) films using ionic liquids, *J. Mater. Chem.* 18 (2008) 5354–5358, <https://doi.org/10.1039/B808723G>.
- [6] A.M. Nardes, R.A.J. Janssen, M. Kemerink, A Morphological Model for the Solvent-Enhanced Conductivity of PEDOT:PSS Thin Films, *Adv. Funct. Mater.* 18 (6) (2008) 865–871, <https://doi.org/10.1002/adfm.200700796>.
- [7] K. Lim, S. Jung, S. Lee, J. Heo, J. Park, J.-W. Kang, Y.-C. Kang, D.-G. Kim, The enhancement of electrical and optical properties of PEDOT:PSS using one-step dynamic etching for flexible application, *Org. Electron.* 15 (2014) 1849–1855, <https://doi.org/10.1016/j.orgel.2014.04.014>.
- [8] A.M. Higgins, S.J. Martin, P.C. Jukes, M. Geoghegan, R.A.L. Jones, S. Langridge, R. Cubitt, S. Kirchmeyer, A. Wehrum, I. Grizzi, Interfacial structure in semiconducting polymer devices, *J. Mater. Chem.* 13 (11) (2003) 2814–2818, <https://doi.org/10.1039/B304990F>.
- [9] Y. Shi, Y. Zhou, R. Shen, F. Liu, Y. Zhou, Solution-based synthesis of PEDOTPSS films with electrical conductivity over 6300 S/cm, *J. Ind. Eng. Chem.* 101 (2021) 414–422, <https://doi.org/10.1016/j.jiec.2021.05.036>.
- [10] D. Yun, J. Jung, Y. Sung, H. Kim, J. Chung, S. Kim, Y. Kim, S. Heo, K. Kim, Y. Jeong, J. Jang, In-Situ Photoelectron Spectroscopy Study on the Air Degradation of PEDOT: PSS in Terms of Electrical and Thermoelectric Properties, *Adv. Electron. Mater.* 6 (2020) 2000620, <https://doi.org/10.1002/aelm.202000620>.
- [11] A. Schultheiss, M. Gueye, A. Carella, A. Benayad, S. Pouget, J. Faure-Vincent, R. Demadrille, A. Revaux, J.-P. Simonato, Insight into the degradation mechanisms of highly conductive poly (3, 4-ethylenedioxythiophene) thin films, *ACS Applied Polymer Materials.* 2 (7) (2020) 2686–2695, <https://doi.org/10.1021/acsapm.0c00301>.
- [12] E. Bae, Y. Kang, K. Jang, S. Cho, Enhancement of Thermoelectric Properties of PEDOT:PSS and Tellurium-PEDOT:PSS Hybrid Composites by Simple Chemical Treatment, *Sci. Rep.* 6 (2016) 18805, <https://doi.org/10.1038/srep18805>.
- [13] H. Ling, J. Lu, S. Phua, H. Liu, L. Liu, Y. Huang, D. Mandler, P.S. Lee, X. Lu, One-pot sequential electrochemical deposition of multilayer poly(3,4-ethylenedioxythiophene):poly(4-styrenesulfonic acid)/tungsten trioxide hybrid films and their enhanced electrochromic properties, *J. Mater. Chem. A.* 2 (2014) 2708–2717, <https://doi.org/10.1039/C3TA14781A>.
- [14] R. Colucci, G. Faria, L. Santos, G. Gozzi, On the charge transport mechanism of cross-linked PEDOT:PSS films, *J. Mater. Sci.: Mater. Electron.* 30 (2019) 16864–16872, <https://doi.org/10.1007/s10854-019-01474-y>.
- [15] J. Park, T. Kang, B. Kim, H. Lee, H. Choi, J. Yook, Real-time Humidity Sensor Based on Microwave Resonator Coupled with PEDOT:PSS Conducting Polymer Film, *Sci Rep.* 8 (2018) 439–446, <https://doi.org/10.1038/s41598-017-18979-3>.
- [16] J. Ouyang, C. Chu, F. Chen, Q. Xu, Y. Yang, High-Conductivity Poly(3,4-Ethylenedioxythiophene):Poly(styrenesulfonate) Film and Its Application in Polymer Optoelectronic Devices, *Adv. Funct. Mater.* 15 (2005) 203–208, <https://doi.org/10.1002/adfm.20040016>.
- [17] H. Seo, M. Son, N. Itagaki, K. Koga, M. Shiratani, Polymer counter electrode of poly (3,4-ethylenedioxythiophene):Poly(4-styrenesulfonate) containing TiO₂ nanoparticles for dye-sensitized solar cells, *J. Power Sources.* 307 (2016) 25–30, <https://doi.org/10.1016/j.jpowsour.2015.12.112>.